Milad Mesbahi

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Education & Technical Projects

University of California (B.S.M.E), Los Angeles
October 2020 - June 2024

Autonomy/Robotics

- ✓ Engineered a wireless-controlled rover capable of autonomous navigation and environmental interaction, integrating embedded systems and control mechanisms.
- ✓ Led the design and construction of a one-pound combat robots using CAD, FEA, and electronics, securing 2nd place in a competitive battle bot event.

Computer-Aided Design

- ✓ Conceptualized and designed a six-degree-of-freedom robotic arm for factory lines, comprising over fifty parts.

Computer Science Background

- ✓ Developed a C++ two-player video game and a scalable movie recommendation system for 20k movies and 100k users.

Physics

- ✓ Innovated an Arduino-based guitar tuner applying Fourier transforms for accurate note detection and tuning via microphone analysis.

Fluid Dynamics

- ✓ Authored a detailed review on dragonfly flight, emphasizing wing biomechanics and agility, leveraging mathematical and Computational Fluid Dynamics models.

Technical Skills

- CAD/FEA/FEM(Solidworks/NX)
- Languages: C++, MatLab, Python, HTML, JS
- Standard Shop Tool, Electronic Bench, and Mfg. Proficiency
- MSSuite, Mac Software Bundle, & Google products

Non-Technical Skills

- Self-Starter
- Collaborative Mindset, Goal Oriented
- Organized (President of School Social Organization)
- Foreign Languages: Spanish and Farsi

Recommendations

- *Associate Director Sales, S&P Global Mobility*

  "Milad is very fortunate to have the special combination of being technically proficient, beautifully blended with strong interpersonal skills.

  These two blended skillsets alone make him a valuable resource for any hiring company, or anyone lucky enough to be his peer, but these assets are then backed by someone who is a self-starter, thrives to innovate, and can breathe through difficult circumstances.

  I could write more, but I think the key takeaway here is that you should meet this young professional and realize the true impact potential that is available."

- *Head of Growth & Founding Team, BeReal*

  "Milad is a standout in creativity and intelligence. His ideas and work ethic transformed our growth strategy, showing a rare blend of innovation and professionalism.

  I've seen Milad excel and inspire, and I'm confident he is destined for remarkable achievements. He is truly a one-of-a-kind talent and person, an asset to any team."

Professional Experience

Mechanical Engineering Internship (Systems Testing)

Jet Propulsion Laboratory - NASA
Pasadena / June 2023 - September 2023

- ✓ Led a research project through the mission assurance and reliability division utilizing a novel new 3D chainmail, achieving a -20dB reduction in transmissibility at certain decisive frequencies and-30dB over 800hz
- ✓ Conducted comprehensive vibrational analysis, post signal processing (MatLab) Finite Element Method (FEM) simulations, demonstrating chainmail's efficacy in aerospace damping.

Mechanical Engineering Researcher (Undergraduate)

Thermoscience Laboratory - UCLA
Los Angeles / October 2022 - Present

- ✓ Lead research at UCLA's Thermosciences Lab on liquid desiccants in fluid dynamics, leading rigorous experiments that I integrate with simulation data.
- ✓ Assisted development of MATLAB algorithms for automated analysis of liquid desiccants and tension-curvature relations, advancing the lab's research towards publication.
- ✓ Enhanced experimental accuracy by designing and engineering custom parts and exploring various new avenues supported by my PI.

Co-Founder of Social Networking Startup

Me and You Always (MAYA) - UCLA
Los Angeles / October 2022 - Present

- ✓ Co-founded a social networking platform focusing on authentic, local community interactions, learning web development, UI/UX design, and marketing from scratch.
- ✓ Directed strategic and creative entrepreneurship processes, including market research, user feedback sessions, and product development.
- ✓ Launched in June 2023, growing the user base to over 3,000 UCLA students with ongoing enhancements and expansion plans in 2024.

Marketing & Growth Internship

Campus Launch Program - BeReal.
Los Angeles / September 2021 – June 2022

- ✓ Spearheaded BeReal's marketing and expansion in the Los Angeles area, targeting college and local markets.
- ✓ Implemented unique, guerilla marketing techniques that allowed Los Angeles the app's fastest-growing location at the time.
- ✓ Engaged and accessed my extensive college network and helped produce and compile user feedback for feature optimization.